PROJECT: Student & Course Management System

PART 1: Database Creation Questions

- Create a database named student_management.
- Create a students table with student_id, first_name, last_name, email, date_of_birth, gender.
- Create an instructors table with instructor_id, instructor_name, email.
- Create a courses table with course_id, course_name, instructor_id.
- Create an enrollments table with enrollment_id, student_id, course_id, enrollment_date.
- Create a grades table with grade_id, enrollment_id, score.

PART 2: Data Insertion Questions

- Insert at least 20 students.
- Insert at least 5 instructors.
- Insert at least 8 courses.
- Insert at least 40 enrollments.
- Insert at least 40 grades.
- Ensure each student is enrolled in at least one course.
- Ensure each course has at least 3 students.
- Ensure each enrollment has one grade.

PART 3: SQL Query Questions

- Display all students with their enrolled courses.
- Show each instructor and the courses they teach.
- List students who scored above 80.
- Find the average score for each course.
- Find the total number of students in each course.
- List the top 5 highest scoring students.
- Show students who failed (score below 50).
- Display courses with more than 5 students enrolled.
- Find the youngest student.
- Find the oldest student.
- Show students who are not enrolled in any course.
- Count how many students each instructor is teaching.

PART 4: Intermediate SQL Questions

- Rank students by their average score.
- Find the course with the highest average score.
- Find the instructor whose students perform best on average.
- Display students who are enrolled in more than 2 courses.
- Find students who have the same first name but different IDs.
- Show the percentage of students who passed each course.
- List courses with no failing students.
- Find students whose average score is above the overall average score.

PART 5: Business & Reporting Questions

- Generate a report showing student name, course name, instructor name, and score.
- Generate a summary report of total students, total courses, total instructors, and average score.
- Identify students who need academic support (score below 50).
- Recommend the best performing course based on average score.
- Recommend the best performing instructor based on student results.

FINAL DELIVERABLE

- Submit SQL file for table creation.
- Submit SQL file for data insertion.
- Submit SQL file containing answers to all questions.
- Submit a short presentation explaining database design, key queries, and insights discovered.